SKYLARK BUSINESS INTELLIGENCE AGENT
DECISION LOG

Assignment: Skylark Drones – Business Intelligence Agent
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Purpose: Document key assumptions, technical decisions, trade-offs, and interpretation of the assignment requirements.

1. Objective

The objective was to build a working Business Intelligence prototype that connects operational data from Monday.com with an executive dashboard and a conversational AI assistant.

The solution was designed to answer practical management questions around **pipeline, sector concentration, financial position, receivables, deal risks, and data quality**, while keeping numerical business outputs grounded in the underlying data.

2. Key Assumptions

Business Data

- The provided Deal Funnel and Work Order Tracker datasets represent the primary source of business information.
- Deal values, probabilities, sectors, billing, collection and work-order fields are treated as the available source data for analysis.
- Missing fields are treated as **data-quality limitations**, rather than being inferred or replaced with assumptions.
- Monetary values are presented in **INR**, using business-friendly formats such as ₹ Cr and ₹ L.

Leadership Updates

I interpreted **"leadership updates"** as concise information that helps management understand:

1. What is currently happening in the business.
2. Where the largest financial or pipeline exposure exists.
3. Which deals require attention.
4. What data-quality limitations may affect decision-making.
5. What practical actions management should consider.

Therefore, the AI assistant's management briefing focuses on **observations, priority risks and recommended actions**, rather than simply returning raw dashboard metrics.

3. Architecture & Technology Decisions

React + FastAPI

I chose **React** for the frontend and **FastAPI** for the backend.

Why:

- Fast to develop within the assignment's time constraint.
- Clear separation between presentation and business logic.
- FastAPI provides simple REST endpoints and automatic API documentation.
- React supports a responsive, interactive dashboard.

Modular Analytics Layer

Business calculations were separated into dedicated analytics modules for:

- Pipeline
- Finance
- Operations
- Risk
- Data Quality

This avoids embedding business logic directly inside API endpoints or UI components and makes the system easier to extend.

Monday.com API Integration

I used a dedicated Monday.com client to retrieve board data and kept the integration separate from the analytics layer.

This allows the source system to be changed or extended without rewriting the dashboard's business calculations.

4. AI Architecture Decision

I chose a **hybrid AI architecture** instead of allowing the LLM to directly calculate business metrics.

The flow is:

```
User Question
   ↓
Query Routing
   ↓
Deterministic BI Tools
   │
   ├── Pipeline
   ├── Sector
   ├── Finance
   ├── Receivables
   ├── Risk
   └── Data Quality

Management / Ambiguous Query
   ↓
LLM Intent / Synthesis
   ↓
Verified Analytics Facts
   ↓
Grounded Response
```

The application uses **Qwen2.5 1.5B through Ollama** for natural-language understanding and management-level response generation.

Why this approach?

A pure LLM approach could produce plausible but incorrect financial figures. Instead, Python performs the actual calculations and retrieves business facts. The LLM is then given verified values for interpretation and presentation.

For management briefings, a **verified fact sheet** is constructed from the analytics layer before the LLM generates the response. The prompt explicitly prevents changing numbers, inventing trends, inventing deals or introducing unsupported recommendations.

This provides a practical balance between **AI usability and numerical reliability**.

5. Key Trade-offs

Deterministic Routing vs Fully Autonomous Agent

I prioritized deterministic routing for common BI questions.

Trade-off:

The system is less autonomous than a fully agentic tool-calling architecture, but responses to important financial and operational questions are more predictable and easier to validate.

Local LLM vs Hosted LLM

The prototype uses Qwen2.5 1.5B through Ollama.

Why:

It provided a lightweight model suitable for rapid prototyping and avoided making the core business logic dependent on a large external model.

Trade-off:

A larger hosted model could provide stronger language understanding and more sophisticated reasoning, but would introduce additional infrastructure, cost and dependency considerations.

Predefined Analytics vs Free-form Analysis

Core metrics are calculated using predefined business logic.

Trade-off:

This limits arbitrary analytical questions, but ensures consistent definitions for metrics such as pipeline, weighted pipeline, billing, collections and receivables.

6. Error Handling & Data Quality

The backend uses exception handling around external data access and exposes appropriate API errors instead of allowing failures to crash the application silently.

Data-quality analysis was also treated as a first-class feature.

Missing values such as:

- Deal values
- Probabilities
- Sectors
- Work-order amounts
- Billing information
- Collection information

are surfaced to the user.

This is important because incomplete source data can directly affect the reliability of downstream business analysis.

7. What I Would Do Differently With More Time

With additional development time, I would:

- Implement a more fully autonomous tool-calling agent.
- Add authentication and role-based access.
- Add persistent conversation history.
- Add automated tests for analytics calculations and API endpoints.
- Add stronger schema validation for Monday.com responses.
- Introduce caching and background refresh for larger datasets.
- Add historical trend analysis instead of only current-state intelligence.
- Improve monitoring, logging and production observability.
- Use a stronger production-grade LLM where appropriate.
- Add richer executive reporting and export functionality.

8. Final Design Rationale

The main design principle was "AI should enhance business intelligence, not replace the source of truth."

The application therefore separates:

Data → Analytics → Verified Business Facts → AI Interpretation → Executive Insight

This makes the prototype useful as an executive decision-support system